Beyond Mapping: A Theory-Grounded Framework for Evaluating Analogical Transfer in Language Models

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Abstract

Language models can benefit from analogies without reliably transferring their relational structure. Additional explanation and answer cues can improve accuracy, while a net gain can conceal regressions. We introduce a theory-grounded diagnostic framework that operationalizes analogical reasoning through retrieval, mapping, schema induction, and adaptation. Explicit stage artifacts and information boundaries allow unsuccessful paths to be inspected and rerun after reference-informed upstream corrections. Three complementary evaluations link outcome measurement to process diagnosis. Scientific question-answering controls show that analogy gains need not exceed explanation or random-example effects. A factorial narrative experiment finds structure-dependent source-answer benefits, including transfer interactions of 14.24 and 12.94 percentage points for Llama-3.1-8B and 70B. In the diagnostic workflow, all initial GPQA failures occur at retrieval, yet 75.34% of that failure cohort still reaches inadequate adaptation after upstream corrections. This separates the first visible barrier from the remaining difficulty. The framework enables more precise evaluation of helpful and harmful analogy use while distinguishing assisted continuation from independent model success.

1 Introduction

An analogy is useful when a relation learned in one situation can guide reasoning in another. A source may describe an army divided across several routes, while a target asks how to concentrate radiation on a tumor without damaging surrounding tissue. The useful correspondence concerns distributed forces, safe paths, and coordinated convergence, rather than shared words or objects (Gick and Holyoak, 1980). To solve the target, a reasoner must obtain an appropriate source, preserve its relational structure, and adapt the resulting principle to new constraints.

Language models can produce plausible analogies and sometimes answer more accurately after receiving them (Yasunaga et al., 2024; Yuan et al., 2024). However, random examples can also improve performance (Qin et al., 2025). An analogy prompt changes several things at once: the amount of explanation, the information available, the solution cues, and the requested reasoning procedure. An accuracy gain therefore leaves open what the model transferred. Even when the source is useful, a wrong target answer does not reveal which operation failed.

We argue that evaluating analogical reasoning requires linking three questions: *What changed in the answer? Does the benefit depend on structural correspondence? Where does the attempt to use that correspondence break down?* Final-answer accuracy addresses the first only partially. It can hide regressions behind a net improvement. Explicit references to an example do not answer the second, because the model can mention a source while misusing or rejecting it. A single mapping score cannot answer the third, because source access, abstraction, and target application can fail separately.

We introduce a theory-grounded diagnostic framework that operationalizes analogy through **retrieval**, **mapping**, **schema induction**, and **adaptation**. The stages produce explicit artifacts under restricted information interfaces. On unsuccessful trajectories, we locate the first inadequate artifact, replace it with a reference-informed correction, and ask the same base model to continue. Repeating this procedure exposes downstream difficulties that were hidden by an earlier failure. The contribution is this operationalization of established theoretical distinctions, rather than the claim that analogy has exactly four universal steps.

Three complementary evaluations establish the framework's empirical setting. Controlled scientific question answering shows why added explana-

tion, unrelated examples, and answer regressions matter for attributing analogy effects. A factorial narrative experiment tests whether a source answer helps selectively when the source has an intended structural correspondence. Finally, interventions in the four-stage workflow identify failures in scientific and narrative reasoning. Larger Llama and Qwen models show structure-dependent source-answer benefits, while the diagnostic study reveals that initial retrieval failures can conceal persistent adaptation failures.

Our contributions are (1) an executable framework connecting cognitive distinctions to auditable outputs and cumulative continuation tests; (2) an evaluation account separating net gains, helpful and harmful answer changes, textual mention, and structure-dependent benefits; and (3) empirical evidence that successful source use is possible under a controlled protocol, but supplying useful upstream information does not reliably resolve target adaptation. These findings motivate evaluating the whole source-to-target process, beyond mapping or accuracy alone.

2 A Theory-Grounded Diagnostic Framework

2.1 From cognitive distinctions to stage contracts

Structure-mapping theory treats analogy as correspondence between relational systems rather than isolated attributes (Gentner, 1983). Systematicity favors connected, especially causal, relations over disconnected local matches (Clement and Gentner, 1991); computational structure mapping makes consistency constraints explicit (Falkenhainer et al., 1989). However, the availability of a source and the soundness of an inference from it are separable (Gentner et al., 1993; Holyoak and Koh, 1987). Comparing sources can yield a schema that abstracts the solution principle and its applicability conditions (Gick and Holyoak, 1983; Gentner et al., 2003). Applying that schema remains a further problem: target-specific constraints can defeat transfer even when an appropriate source is accessible (Novick and Holyoak, 1991).

We translate these distinctions into four externally controlled calls. Let x denote the target, r the solved sources, m their mapping, s a schema, and a the adapted solution. The permitted information

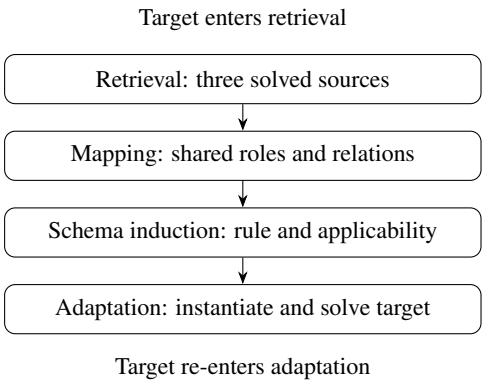


Figure 1: An observable source-to-target workflow. Mapping compares sources; schema induction abstracts their mapped structure; adaptation reconnects the schema to the target. Each stage receives a restricted artifact rather than the entire preceding conversation.

flow is

$$r = R(x), \quad m = M(r), \\ s = S(m), \quad a = A(x, s). \tag{1}$$

Each call starts fresh; only the permitted recorded artifact passes forward. Figure 1 and Table 1 describe the interfaces. The natural workflow never receives the target reference answer.

Retrieval. This stage obtains three solved source problems. In our implementation, the model can generate them; "retrieval" names their functional role, not an observation of access to internal memory. Topical resemblance is insufficient: a source must preserve relations useful for solving the target, and its own explanation must be defensible.

Mapping. The model compares the retrieved sources with one another, recording shared roles, relations, useful examples, and consequential differences. It does not see the target. Consistency and connected relations matter more than a list of similar nouns. Alignable differences help distinguish incidental changes from discrepancies that invalidate transfer (Gentner and Markman, 1997).

Schema induction. The model sees only the recorded mapping. It must express an abstract solution rule together with conditions and limits. A schema should be general enough to cross domains while retaining the operations needed for a solution. The restriction tests whether the mapping artifact supports abstraction without rereading the original narratives.

Stage	Permitted input	Required artifact and adequacy criterion
Retrieval	Target and choices	Three solved sources with answers and explanations; sources must support a solution-relevant relational structure.
Mapping	Retrieval artifact only	Corresponding roles and relations across sources, source suitability, and material differences; correspondences must be consistent and informative.
Schema induction	Mapping artifact only	Abstract solution rule, roles, relations, applicability conditions, and limits; enough structure must remain to guide a new solution.
Adaptation	Target and necessary schema content	Instantiated roles, target-constraint checks, necessary adjustments, and a final answer supported by target evidence.

Table 1: Stage contracts make theoretical distinctions operational. The raw target is withheld from the two middle stages, and raw sources are withheld from adaptation.

Adaptation. The target re-enters with the schema, but without raw sources or the mapping. The model instantiates roles, adjusts domain-specific details, checks constraints, and answers. This is where the shared structure is connected to the actual target rather than merely described.

In the convergence example, the sources must retain that concentrated force is unsafe along a path, weak forces alone are insufficient, and coordinated forces combine at the goal. The mapping aligns this relational system; the schema states the conditions under which convergence works; adaptation checks whether the target satisfies them. "Organize resources better" is too general, while copying a source-specific object or answer position fails to instantiate the relation.

These boundaries define an experimental source-comparison-to-schema-to-target path. They may impose costs absent from ordinary prompting, and they are not a universal temporal model of cognition. Their value is to specify which artifact supports each next operation, making failures inspectable and replacements interpretable.

2.2 Diagnosing first failure and downstream burden

For each naturally unsuccessful trajectory, we inspect artifacts in stage order and record the first one inadequate for its assigned function. The label identifies the first observable failure in this workflow, not the unique hidden cause of the answer. An inadequate retrieval can mask whether the model would also fail to map, abstract, or adapt a good source.

We then replace only that artifact with a corrected one, retain earlier artifacts, and rerun downstream calls with the same base model. If the answer becomes correct, the continuation is rescued under the correction. Otherwise, the new path is inspected and the next inadequate artifact is corrected while prior corrections remain. This yields two distinct observations: the *initial first-failure distribution*, and the *burden revealed over cumulative repairs*. The latter can count one question at several stages.

GPT-6 Sol authors corrected artifacts with access to the target reference answer, an *open-oracle* construction setting. Upstream corrections must be defensible under their stage contract and must not explicitly disclose the target answer. The base model continuing the workflow never receives the reference. This setting tests whether it can exploit solution-relevant upstream guidance; it does not measure autonomous source generation or an isolated causal effect of each stage.

Adaptation requires special treatment. A corrected adaptation artifact already includes the final answer and leaves no downstream model call. Agreement with the reference after that replacement is therefore not a rescue by the base model. We count genuine continuation only after retrieval, mapping, or schema corrections, and keep final oracle outputs separate. This distinction prevents an answer supplied by the correction procedure from being mistaken for successful transfer by the solver.

3 Evaluation Design

The framework motivates three evaluation questions: do apparent analogy gains survive appropriate controls; do source solutions help selectively through intended correspondence; and which stage difficulties remain after upstream repair? The first two experiments examine outcomes under different source conditions. The third evaluates the imposed four-stage workflow. They provide complementary evidence, rather than a single benchmark score or one identical protocol.

3.1 Tasks and response accounting

GPQA Main contains 448 graduate-level science questions (Rein et al., 2024), with 78 biology, 183 chemistry, and 187 physics items in the local version. We use Main rather than the 198-item Diamond subset. MuSR supplies murder mysteries, object placements, and team allocation (Sprague et al., 2024). These tasks respectively require combining clues, tracking locations and agents' knowledge, and satisfying assignment constraints. GPQA often requires external scientific knowledge; MuSR provides much of the relevant evidence in its narratives, facilitating sources with changed entities but preserved solution relations.

Accuracy uses the reference choice. A recorded failed response or unextractable final answer is incorrect. Matched contrasts align identifiers, not file order. A missing question record is distinct from a recorded invalid response. Repeated calls to a question characterize execution variability, not additional independent questions. We interpret numerical patterns descriptively and distinguish run standard deviation from inferential uncertainty.

3.2 Controls for apparent analogy benefits

The scientific study compares concept-focused teacher analogies with explanations, random analogies, and solver-generated reasoning. DeepSeek V4 Flash creates the teacher content. Solvers are Llama-3.1-8B-Instruct, Qwen3-8B, Mistral-7B-Instruct-v0.3, and Gemma-4-31B-it. CoT and self-analogy provide strategies without an external teacher example (Wei et al., 2022; Kojima et al., 2022).

We compare a 300-word analogy with a 600-word analogy, and one, two, or three examples under a nominal 600-word total budget. Non-analogical explanations have 300- or 600-word budgets. Random examples come from the same or a different scientific domain. Each comparison has a specific purpose: length probes additional text; explanations probe whether benefits require analogical presentation; random sources probe the intended relation; and multiple examples probe the opportunity for comparative abstraction. The last comparison does not directly establish that a schema was induced.

A net gain can conceal both improvements and regressions. On the same items, with baseline b and condition c, we measure

$$p_c - p_b = H_{b\to c} - D_{b\to c}, \tag{2}$$

where H is the fraction changing from incorrect to correct and D the fraction changing from correct to incorrect. Both use all shared items as denominator. We separately record explicit reference to the analogy, example, or teacher, including references that reject it. Mention and correctness are distinct observations; neither directly identifies valid structural use.

Llama, Qwen, and Mistral have 448 retained responses in each reported condition. Gemma has 435–447. Its overall accuracies can reflect changes in question composition; paired contrasts use shared identifiers. Appendix A gives the counts and condition definitions.

3.3 A factorial test of source-solution transfer

For each selected MuSR target, GPT-5.6-Sol constructs a source with changed entities and context but intended shared solution relations. The generator sees the target reference answer; the solver does not. Sources contain a narrative, question, choices, and a separate correct source-answer field. Construction is checked for completeness, intended structural suitability, and absence of explicit target-answer disclosure, with manual review of difficult examples and a small model-based assessment rather than exhaustive human validation.

The experiment uses the 103 completed records available at execution: 61 murder mysteries, 21 object placements, and 21 team-allocation questions. This is neither full MuSR nor a documented representative random sample. Random controls reassign the same sources one-to-one among targets with equal choice counts, without self-assignment. Assignment reduces same-story and same-task matches where possible. Four five-choice object-placement items retain same-story controls because alternatives are unavailable. Random reassignment removes the designed correspondence, not every incidental useful relation.

We cross source type with source-answer availability: A is corresponding/withheld; B corresponding/provided; C random/withheld; and D random/provided. Withholding removes only the answer field, retaining source narrative, question, and choices. The target, neutral "additional example" label, instruction, temperature zero, and 2,048-token output ceiling are fixed. Selective source-answer benefit is

$$\Delta_{\text{transfer}} = (p_B - p_A) - (p_D - p_C). \tag{3}$$

A positive interaction supports a greater answer

benefit under intended correspondence. We report every cell because an interaction may combine improvement with damage to a control. Zero interaction can coexist with a useful corresponding source.

Llama-3.2-1B and 3B and Llama-3.1-8B and 70B Instruct have three runs per cell. Qwen3-14B and 32B have two. Provider and quantization differences prevent interpreting the comparison as a pure effect of model size. Random controls permute the same sources, making corpus-wide narrative lengths identical (mean 318.99 words), but not necessarily matching each pair or task. Source-choice shuffling updates answer indices. We compare B on matching versus differing source/target answer positions to probe index following; the strata contain different questions, not controlled reorderings of the same target.

3.4 Stage diagnosis and continuation tests

We apply the workflow in Section 2 to Llama-3.1-8B-Instruct on 100 GPQA Main questions and 100 MuSR murder mysteries. Three solved sources are requested per target. We diagnose naturally unsuccessful workflows and perform the cumulative upstream repairs of Section 2.2. The question is whether corrected intermediate guidance enables the base model to finish, and which inadequate stages emerge when it does not.

Human review covers 30% of questions and associated outputs for logical coherence and correctness. The remaining cases have no human review; reference-informed correction is not equivalent to exhaustive validation. We report natural accuracy, the initial failure cohort, first-failure assignments, cumulative rescue, and overlapping stage burdens separately. All counts use the actual 100-item cohorts. Doubled display counts in some retained thesis graphics add no observations and are not used here.

4 Results and Analysis

4.1 The first failure can conceal the main remaining obstacle

The diagnostic framework answers 27/100 GPQA questions and 57/100 murder mysteries correctly without corrections. The 73 GPQA failures are all first diagnosed at retrieval. The 43 murder-mystery failures begin at retrieval in 19 cases, mapping in 3, and adaptation in 21; none begins at schema induction. These are first-failure assignments within unsuccessful workflows, not unconditional stage error rates.

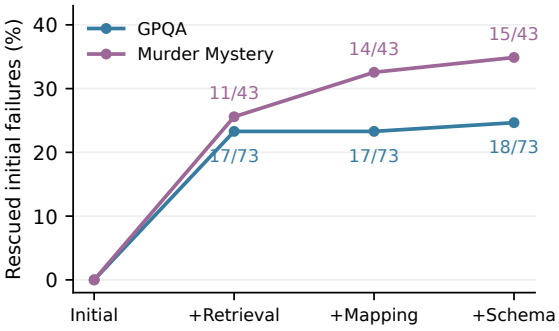


Figure 2: Cumulative rescue of initially failed workflows after upstream corrections. Counts use the actual cohorts: 73 GPQA and 43 murder-mystery failures. Final gold adaptation is excluded because it supplies the answer rather than testing downstream base-model continuation.

A first-failure label alone suggests that GPQA needs better sources and murder mysteries need better target application. Cumulative repair tests what this account leaves out. After retrieval correction, the same base model solves 17 of the 73 GPQA failures (23.29%). Mapping correction does not add rescues, and schema correction adds one, bringing the total to 18/73 (24.66%). For murder mysteries, the cumulative totals are 11, 14, and 15 of 43 failures after retrieval, mapping, and schema correction (25.58%, 32.56%, and 34.88%). Figure 2 follows the same initial failure cohorts.

Failure migration and overlapping burden. After upstream corrections, adaptation remains inadequate for 55/73 GPQA failures (75.34%) and 28/43 murder mysteries (65.12%). GPQA also reveals mapping problems in 26/73 and schema problems in 15/73 over the correction history; murder mysteries reveal mapping problems in 6/43 and schema problems in 1/43. These burdens overlap because one trajectory can need several repairs. Appendix B separates mutually exclusive current states from overlapping cumulative burdens.

The distinction changes the interpretation of the initial diagnosis. Retrieval is the first barrier for every failed GPQA trajectory, but it is not the only barrier: most paths still fail after upstream guidance is corrected. In particular, moving a failure from retrieval to adaptation does not by itself indicate that the retrieval correction damaged the path. It can expose an operation that the original path never reached with an adequate input. Likewise, mapping correction can remove a visible deficiency without

increasing final accuracy, because the schema or adaptation remains inadequate.

Rescue requires base-model continuation. Natural successes plus genuine continuation after upstream corrections give 45/100 correct GPQA answers and 72/100 correct murder mysteries. These are assisted workflow outcomes. Replacing adaptation with a reference-informed final answer is excluded: that intervention supplies the result instead of testing the solver's ability to finish. Reporting it as model rescue would erase precisely the downstream difficulty that the framework identifies.

The domain contrast is compatible with the task demands. Scientific questions require external knowledge to understand the target and obtain useful sources. Murder mysteries provide many relevant facts but require instantiating relations under particular story constraints. These results do not identify hidden mechanisms; they show why observable source adequacy and target adaptation need separate assessment.

4.2 Solved sources can support structure-dependent transfer

To establish that sources can help through more than their mere presence, we examine the factorial source-answer test in Table 2. Llama-1B gains 5.50 points when either the corresponding or random source answer is provided, yielding zero mean interaction. Llama-3B gains about 13.92 points from corresponding source content in both answer settings, but the answer adds the same 2.91 points in either source type. Its zero interaction therefore does not mean the corresponding source is useless; it means the observed answer benefit is not selective.

Llama-8B and 70B show a different pattern. The 8B model gains 13.59 points from the corresponding answer and changes by -0.65 points with the random answer, giving a 14.24-point interaction. The 70B model gains 12.94 points from the corresponding answer and none from the random answer. These results support structure-dependent use of source answers under the tested protocol. They do not show that every correct answer used the source or that transfer grows monotonically with model size.

The pattern also appears outside the Llama family. Qwen-14B gains 9.22 points from the corresponding answer, with no change in the random-answer mean. Qwen-32B combines a 5.34-point corresponding-answer gain with a 7.28-point random-answer loss, giving a 12.62-point interaction. Its larger interaction does not mean a larger benefit from the valid guide: part of the difference comes from damage in the control. The individual cells are essential to the interpretation.

Transferring an answer role is different from copying an index. Llama-1B achieves 88.24% accuracy when source and target correct-answer positions match and only 0.64% when they differ. Together with equal answer gains for corresponding and random sources, this suggests predominant index following. Llama-8B and 70B retain 57.05% and 65.38% accuracy in the different-position stratum, compared with 74.51% and 79.08% in the matching stratum. Their performance is compatible with more than index copying, although position sensitivity persists. Since the strata contain different questions, this check supports rather than independently establishes the factorial interpretation.

Thus, there is positive evidence for selective source use in several evaluated models, alongside substantially different behaviors in smaller models. The framework is useful in this setting because a capacity demonstrated under a supplied source does not guarantee that a model can obtain, abstract, and apply an adequate source on its own.

4.3 Aggregate gains and textual mention are insufficient measures

The scientific controls explain why the stronger attribution test and stage diagnosis are needed. In Table 3, the concept-focused 300-word analogy does not outperform both random controls for any solver. Llama performs slightly better with a same-domain random source, Qwen matches it, and Mistral trails both random controls. Gemma lies between its controls, with the additional caveat of differing retained sets. The intended example relation does not yield a general advantage in final accuracy.

The comparison determines the claim. Mistral's teacher analogy raises accuracy from 16.96% under CoT to 27.01%. Yet a 300-word non-analogical explanation reaches 27.68%. A gain against CoT therefore does not establish a benefit requiring analogical presentation. Llama's teacher analogy and explanation both trail its CoT baseline. Qwen's self-analogy reaches 43.30%, compared with 42.19% for CoT and 36.61% for the

Model	A: true/no answer	B: true/answer	C: random/no answer	D: random/answer	Δ_{transfer}
Llama-3.2-1B	38.51 ± 1.48	44.01 ± 0.56	38.51 ± 2.02	44.01 ± 0.56	0.00
Llama-3.2-3B	54.05 ± 1.48	56.96 ± 0.56	40.13 ± 1.12	43.04 ± 1.48	0.00
Llama-3.1-8B	52.10 ± 3.68	65.70 ± 3.41	52.43 ± 1.94	51.78 ± 7.35	+14.24
Llama-3.1-70B	59.22 ± 3.88	72.17 ± 8.76	53.07 ± 2.24	53.07 ± 2.02	+12.94
Qwen3-14B	66.02 ± 4.12	75.24 ± 0.69	60.68 ± 2.06	60.68 ± 3.43	+9.22
Qwen3-32B	68.93 ± 2.75	74.27 ± 0.69	65.53 ± 2.06	58.25 ± 2.75	+12.62

Table 2: Factorial transfer accuracy (%, mean $\pm$ run SD) on the same 103 MuSR targets in every cell. Llama has three runs; Qwen has two. Interactions are in percentage points, computed from unrounded run-level values; rounding displayed cells can produce small discrepancies. SD is not an inferential confidence interval.

Condition	Llama-8B	Qwen-8B	Mistral-7B	Gemma-31B
Analogy, 300 words	30.80	36.61	27.01	65.39
Analogy, 600 words	31.25	36.83	28.79	68.69
Two analogies, 2×300 words	31.92	35.27	30.58	69.00
Three analogies, 3×200 words	31.03	39.73	26.56	69.61
Non-analogical explanation, 300 words	31.03	37.72	27.68	68.25
Non-analogical explanation, 600 words	30.58	36.61	25.89	68.47
Random same-domain analogy, 300 words	31.25	36.61	27.68	64.85
Random different-domain analogy, 300 words	29.69	39.73	28.13	67.12
Chain-of-thought	32.14	42.19	16.96	66.52
Self-analogy	28.79	43.30	22.54	64.06

Table 3: Scientific question-answering accuracy (%) on GPQA Main. Llama, Qwen, and Mistral have 448 retained responses per condition. Gemma has 435–447, so its overall contrasts can also reflect changes in the retained question set. Recorded responses without a final answer are incorrect.

teacher analogy. These are effects of particular assistance and prompting conditions, not interchangeable measures of transfer.

Longer or more numerous examples also fail to provide a uniform benefit. Doubling analogy length adds 0.45 points for Llama and 0.22 for Qwen. At a nominal total of 600 words, Mistral's one-, two-, and three-example accuracies are 28.79%, 30.58%, and 26.56%, while explicit mentions increase from 12.05% to 19.42% and 29.24%. Increased engagement with examples in the output is not direct evidence that a usable schema was induced.

Net improvement can coexist with damage. On the same 448 items, Mistral's 300-word analogy repairs 86 CoT errors (19.20% of all items) while making 41 previously correct answers incorrect (9.15%). The 45-item net gain is 10.04 percentage points. Two 300-word examples repair 105 answers and damage 44. Figure 3 makes both directions visible. These are condition-associated changes: differing instructions and similar changes under random controls prevent assigning every regression to a faulty analogical mapping.

Mention is neither successful transfer nor its necessary trace. Qwen mentions the 300-word example in 31.70% of all responses, compared with

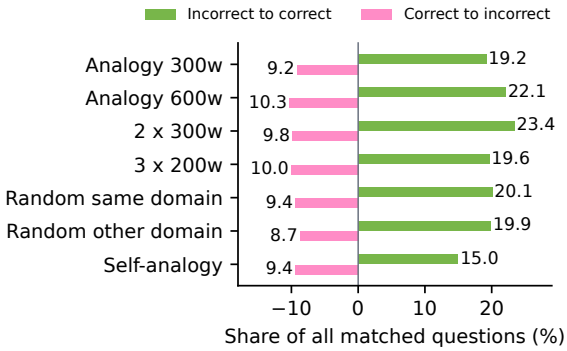


Figure 3: Mistral transitions relative to CoT on the same 448 questions. Improvements and regressions use all shared items as denominator. Their difference is the net accuracy change.

8.31% for Gemma despite Gemma's higher accuracy. Qwen's mention rates within correct and incorrect answers are 31.10% and 32.04%. Explicit reference can accompany either outcome, and absence of reference does not prove that the example had no effect. Diagnosing the actual intermediate artifacts gives a more specific question: did the source provide an adequate relation, was it preserved in the schema, and was it applied under the target constraints?

Together, these evaluations establish a hierarchy

of evidence rather than a single ranking. Outcome changes show practical effects; controls narrow their attribution; selective source-answer gains support transfer under an intended correspondence; and stage interventions reveal what prevents that process from completing. The claim supported by each measurement depends on the information and operations it actually tests.

5 Related Work

Identifying analogies. ARN and AnaloBench evaluate analogy identification in narratives under variations in surface similarity and context (Sourati et al., 2024; Ye et al., 2024). Recognizing a corresponding source is informative, but does not establish that its solution can be adapted to a target. Our framework preserves this distinction by separating source adequacy from later abstraction and application. The theoretical connection follows the broader effort to relate cognitive accounts of analogy to relational understanding in NLP (Petersen et al., 2026); our contribution is an executable diagnostic protocol with restricted interfaces and cumulative repairs.

Using analogies for problem solving. Analogical prompting generates relevant exemplars before solving the target (Yasunaga et al., 2024). SCUA evaluates teacher-generated analogies as explanations of scientific concepts (Yuan et al., 2024). These approaches establish utility under their respective settings, but example quality, additional explanation, and target-solving instructions can change together. We compare concept explanations against specific controls and then construct a factorial source-answer test to distinguish source content benefits from selective solution transfer.

Attributing analogy effects. Qin et al. (2025) show that random self-generated examples can also help, challenging an interpretation based on relevance alone. Our scientific results reinforce this measurement concern while exposing helpful and harmful transitions hidden by a mean. The narrative experiment provides complementary positive evidence by testing answer availability under intended correspondence and random reassignment of the same source corpus. Finally, stage diagnosis moves from whether a source condition helps to which observable operation remains inadequate. We do not claim to originate multi-component accounts of analogy or to infer hidden cognitive processes from generated text.

6 Conclusion

We presented a theory-grounded framework that evaluates analogical reasoning through source retrieval, relational mapping, schema induction, and target adaptation. Explicit artifacts and information boundaries enable cumulative correction tests that separate a first failure from downstream difficulties. The experiments show why final accuracy and textual mention are insufficient, provide positive evidence for structure-dependent source-answer benefits in several models, and reveal adaptation problems that persist after upstream guidance is corrected.

The practical implication is to evaluate what an analogy changes, what makes its solution useful, and what prevents that solution from reaching the target. Better sources alone need not resolve transfer. Future evaluations can vary stage interfaces, expand source validation, and compare matched valid and malformed corrections to refine attribution. The framework provides a testable account of the observable source-to-target process while retaining the distinction between assisted continuation and independent model performance.

Limitations

The framework diagnoses an imposed, text-based workflow. Its intermediates are behavioral artifacts, not direct observations of hidden model computation. Requiring fresh calls and restricting information may introduce difficulties absent from unconstrained solving. The four-stage synthesis is one operationalization of cognitive distinctions; alternative interfaces or orderings may yield different diagnoses.

The scientific control study controls particular alternative explanations but does not isolate every source of condition differences. Instructions differ between teacher-assisted and solver-generated conditions, and nominal word budgets do not guarantee equal token or computational budgets. Gemma's retained question counts vary across conditions. Overall differences for that model can therefore include changes in question composition; matched-item comparisons require shared identifiers. Recorded invalid answers and service failures can also influence accuracy. Llama's recovered scientific-control records combine an initial DeepInfra run with AvalAI recovery of unsuccessful requests.

The factorial transfer experiment uses 103 completed synthetic-source records rather than a representative sample of all MuSR questions. Human source validation is incomplete. Correspondence is an intended and partially reviewed construction property; random reassignment removes the designed correspondence but cannot guarantee the absence of incidental useful relations. The four same-story object-placement controls and task-specific length differences further limit isolation. Global narrative-length matching does not match each pair's information content. The experiment manipulates source-answer *availability*, not correct versus deliberately incorrect source answers.

The model-size comparison also changes Llama generation, provider, and sometimes quantization. The 1B and 3B models use Cloudflare, 8B uses Novita FP8, and 70B uses DeepInfra; Qwen uses DeepInfra FP8. Consequently, these observations do not identify a pure parameter-count effect. Two or three repeated executions describe run variability rather than population uncertainty. Object-placement questions can share stories, and comparisons within answer-position strata do not control item difficulty. We make no significance claim or universal scaling claim from these results.

The diagnostic study covers one reported base model and 100 questions per dataset. Diagnosis focuses on unsuccessful staged outputs and does not estimate unconditional failure rates or spontaneous analogy use. Human review covers 30% of questions and associated outputs, without a reported inter-annotator agreement estimate. GPT-6 Sol authors corrections with access to the reference answer. Even without explicit answer disclosure, answer-informed sources or schemas can carry strong solution guidance. Rescue is therefore an assisted behavioral outcome, conditional on the oracle-informed artifact and the preceding correction history. Final-answer replacements are not independent model successes. These limits motivate reporting natural performance, cumulative upstream rescue, and oracle answer agreement separately.

Ethical Considerations

We study existing scientific and synthetic narrative benchmarks and model-generated sources. Improvements in benchmark performance do not establish reliable scientific or educational advice. An analogy can be persuasive while omitting a decisive constraint; the observed answer regressions reinforce the need to check applicability before deploying analogy-based assistance. Dataset and model access remain subject to their respective licenses and service terms. Generative AI assisted conversion of the research manuscript into English prose and LaTeX; the authors remain responsible for checking the claims, citations, numerical results, and final text. The use of models to generate sources and reference-informed corrections is described in the method.

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A Execution and Evaluation Details

This appendix summarizes the retained experimental protocol. It describes stage tasks and condition construction rather than presenting reconstructed wording as the exact historical prompts.

A.1 Phase-I condition identifiers

The reported 300-word analogy is condition 1, focused on the main scientific concept. Condition 2 uses one 600-word analogy; conditions 3 and 4 use two 300-word and three 200-word analogies. Conditions 5 and 6 are non-analogical explanations of nominal length 300 and 600 words. Conditions 7 and 8 are random same-domain and different-domain analogies. Condition 20 requests CoT and condition 21 requests solver-generated analogy, with a source-generation ceiling of 600 words. The broader study also contains condition 0, an unrestricted-concept analogy; it is not the selected main analogy in Table 3.

Available Gemma response counts for conditions 1, 2, 3, 4, 5, 6, 7, 8, 20, and 21 are respectively 445, 444, 442, 441, 441, 444, 441, 444, 435, and 447. Llama, Qwen, and Mistral have 448 retained responses in each reported condition. Correctness transitions are computed by identifier intersection, with recorded missing final predictions treated as incorrect. A missing question record is distinct from a recorded response lacking a valid answer.

A.2 Phase-II source construction and assignment

Sources contain `analogy`, `question`, `choices`, and `answer` fields. Generation preserves the target choice count, requests a single correct source option, and caps the source narrative below 1,000 words. Source-choice reordering uses seed 20260901; random-control assignment uses seed 20260903, with the source answer updated after reordering. Withholding the answer removes only `answer`. The older exploratory no-solution condition also withheld source choices and is not used as the main factorial comparison.

Mean corresponding/random narrative lengths by task are 371.41/371.41 words for murder mysteries, 189.14/228.76 for object placements, and 296.57/256.95 for team allocation. Their weighted corpus-wide means are both 318.99 words. Four five-choice object-placement questions have same-story random controls. This is a remaining source-construction limitation rather than evidence of complete structural independence.

A.3 Interpreting the answer-position check

In the source-answer condition B, 51 of the 103 questions have matching source and target answer indices and 52 have differing indices. Pooling three Llama runs yields 153 and 156 response events, respectively; it does not yield that many independent questions. The counts of correct matching/differing events are 135/1 for 1B, 93/83 for 3B, 114/89 for 8B, and 121/102 for 70B. These counts reproduce the position-stratified accuracies in Section 4.2.

B Diagnostic Cohorts and State Accounting

Table 4 reports the actual Phase-III counts. Some retained visualization assets display counts multiplied by two. That display transformation preserves percentages but adds no observations; the paper uses the original 100-item dataset cohorts and the 73/43 initial failure cohorts throughout.

The stage burdens accumulated across interventions overlap, unlike the mutually exclusive states in each row. GPQA has 73 paths needing retrieval correction, 26 needing mapping correction, 15 needing schema correction, and 55 reaching inadequate adaptation. For murder mysteries, these counts are 19, 6, 1, and 28. A path can contribute to several of these totals; summing them does not count failed questions.

State	Rescued	R	M	S	A
GPQA: 73 initial failures					
Initial	0	73	0	0	0
After R	17	0	26	7	23
After M	17	0	0	15	41
After S	18	0	0	0	55
Murder mysteries: 43 initial failures					
Initial	0	19	3	0	21
After R	11	0	6	0	26
After M	14	0	0	1	28
After S	15	0	0	0	28

Table 4: Actual cohort counts over cumulative repair. Each row partitions the same initial failures into rescued paths and the current first inadequate stage: retrieval (R), mapping (M), schema (S), or adaptation (A). Final gold adaptation is excluded.

The saved Phase-III base-model identifier is `cf.llama-3.1-8b-instruct-fast`. No completed second-model comparison is included in this manuscript. Natural successes are 27 GPQA and 57 murder-mystery questions; cumulative genuine upstream rescue contributes 18 and 15, respectively. The agreement of reference-informed final outputs is distinct from these base-model continuations.